

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 8/30/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Yes, team was able to meet in lab and over
 - i. All members present in lab 8/15 at 5pm
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. Yes we have met with the team sponsor.
 - i. All members present on 8/13 for Teams call
 - ii. David and Ian present for meeting 8/27 - John out of the country but still in communication
3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. Worked on the presentation as well as went over the core concepts of this project. Did more research on go1 as well as yolo and object detection. Problem with restart process with cameras after once it is killed or video data.
 - b. Microsoft teams was set up as the main form of communication, and structure for the team moving forward. One of the main challenges faced is the fact that our project is more system based and software based so a lot of the tasks are likely to overlap
4. Describe the tasks to be completed the coming week.
 - a. Work on more intricate designs with the robot, such as working on making sure the cameras are working with YOLO correctly and start working on object detection with the robot. Also must find way to restart process on cameras after once it is killed for video data.
 - b. The group will be generating a live feed of footage taken from the dog in order to apply our deep learning model on
 - c. Research will be done on the onboard features of Go1, and their significance/usefulness to the project

Copy the following table from the last week's report (if this is the first report, skip):

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan			
Ian Simmons			
John Sinfuego			

Fill in the following table for this week:

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons	2	0	2
John Sinfuego	1	0	2

Part 2: Individual Progress Reports

Name of the Member: David Uzan

Report Date: 8/30/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the past week**:

1. Task number/name

General Task 1 – I met the other members of the group and project sponsor in lab 408B, where we went over basic robot operations like connecting to different modules and controlling it. We also went over the scope of the project and a road map to complete the tasks at hand.

Personal Week 1 – I adapted code from Ashan's presentation of object detection on cars, which was purely identifying them as they walked into frame, to recreate a base case that instead tracks movement of the objects on an example provided at Ultralytics's github page for its YOLO modules.



2. Short description – these descriptions form the basis of the final project report
 - a. Implementation, testing, results

To start off with the basic understanding of how the robot should identify objects in each environment, I kept the code running in a Google Colab file to allow for different tests to be done asynchronously to allow for easy comparison. The video of people walking shown in the frame

above was tested with 8n and 8x models to account for runtime and resource usage. The 8n model resulted in a faster runtime with decent results as compared to the 8x model which resulted in a longer runtime with exceptional results.

b. Brief and to point

The detection algorithm models used for the initial tests varied in their analysis and with benefits for both models, one for a preferred runtime and the other for preferred accuracy.

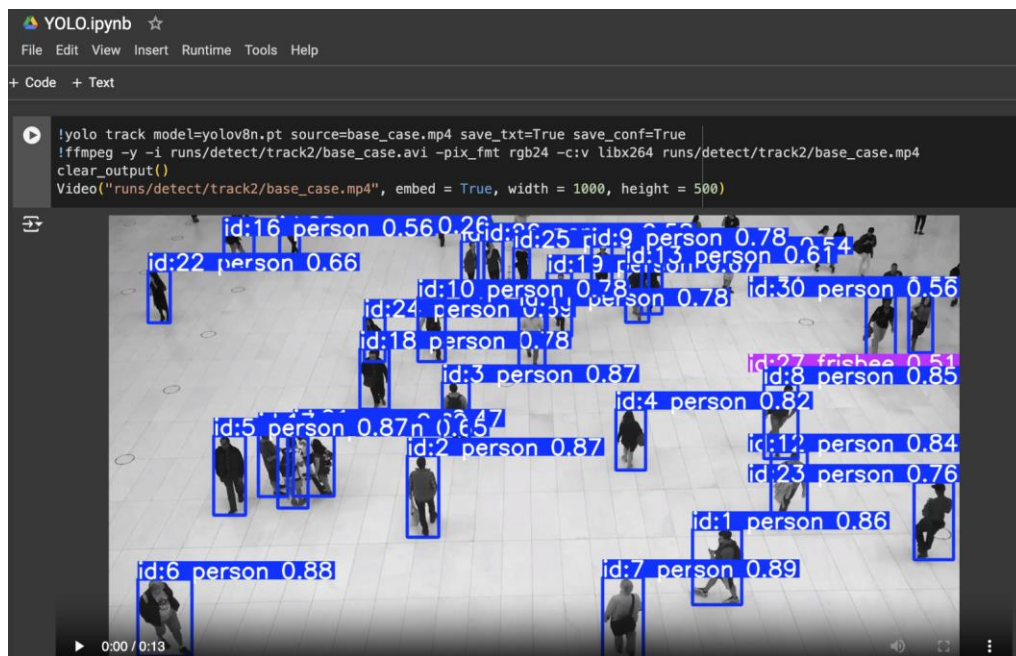
3. Outcome of the task activity

a. How did the task advance your project?

The task advanced the project greatly in our way of understanding how the robot's cameras would identify the objects around it in the field usage.

b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task





For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 2 – Record video data from the robot dog.

Personal Week 2 – Using video recorded by the robot dog, I will apply the detection algorithms to it to find exactly what the robot detects with its cameras.

2. Short description

- a. Plans, resources needed, implementation, results

The group will have the dog walk around the campus and record a variety of objects unique enough to be categorized by the YOLO algorithms. We will bring a change of battery to account for resource usage during recording. Also, we will bring a phone or laptop capable of inputting commands on a separate program used to extract video from the cameras. The results will be shown in the Google Colab.

- b. Main challenges, reasons for any delay in completion

Something that could be a reason for a delay is the GPU usage for running the program and Google Colab's limit on GPU usage for free plans. Previous runs of the tests vary based on file size and frame rate of the video, but a 30 second video took around 10 minutes to test without error.

3. Outcome of the task activity

- a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

N/A, have not started at this time.

Name of the Member: Ian Simmons

Report Date: 8/30/2024

List the following:

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2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the past week**:

1. Week 1, Task 1 (8/30)
 - a. Set up Teams structure and map out beginning steps
 - i. Add tasks bar and begin to populate, add files and folders for organization

The screenshot displays two Jira Kanban boards. The left board, named "General", shows three columns representing weeks. Under "Week 4", there is a task card labeled "Due" with a calendar icon and three status icons (blue, red, red). Under "Week 3", there are two task cards: one for "Establish baseline for use cases and testing" and another for "Search for methods to run YOLOv8 model di". Under "Week 2", there are two task cards: one for "Implement code for video recording on Go1" and another for "Record a video on campus to Go1". The right board, named "lan", also shows three columns. Under "Week 4", there is a task card labeled "09/13" with a blue status icon. Under "Week 3", there are two task cards: one for "Research Jetson Orin Nano kit" and another for "Incorporate the YOLO model into Jetson". Under "Week 2", there are two task cards: one for "09/09" with a blue status icon and another for "09/02" with a blue status icon. At the bottom of each board, there is a summary bar for "Completed tasks" showing 1 task.

 General Posts Files Tasks				
+ New Upload Edit in grid view Share Copy link Sync All Docs				
Documents > General				
	Name	Modified	Modified By	+ Add column
	David	8 hours ago	Ian Simmons	
	Ian	8 hours ago	Ian Simmons	
	John	8 hours ago	Ian Simmons	
	Team Progress Reports	7 hours ago	Ian Simmons	
	Video	6 hours ago	Ian Simmons	
	progress-report-Individual-template-1-1-1....	Tuesday at 5:21 PM	David Uzan	
	progress-report-team-template-1.docx	8 minutes ago	David Uzan	

- b. This will help us stay organized and on track and able to delegate work more effectively. Having a way to see everyone's overall progress will speed up the workflow and improve communication

2. Week 1, Task 2 (8/30)

- a. Begin working on YOLO model on Google Colab
 - i. Originally, the goal was to use Visual Studio Code to incorporate the YOLO model
 1. There are some personal issues that need to be ironed out, Some libraries are not being loaded and it seems as though. The v8 model only natively supports NVIDIA GPUs.
 - ii. Google Colab Is a great way to test out these libraries with minimal errors however, There is a limit to the amount of processing power we can expect to have on Google colab.

For **each** task you are currently working on or plan to work on in the next period:

1. Week 2, Task 1

- a. Research different nanos on Go1
 - i. My goal with this research is to better understand the onboard devices of our dog and hopefully this will generate ways to incorporate our solutions
 - ii. This should help to better incorporate our methods onto the dog to ensure seamless functionality

Name of the Member: John Angelo Sinfuego

Report Date: 8/30/24

List the following:

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2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the past week**:

1. Task number/name

General Task 1 – I met with my team members in the lab in engineering east as well as our sponsors Ashan and Jefferson. We went over the basics and fundamentals of the go1 robot, how it worked, how to control it, its basic architecture, and its technological core and user interface. We then went out to campus and ended up recording some videos with the robot and will implement that into a video where we can upload it using YOLO.

Personal Week 1 – I went and looked over the go1 functional requirements as well as the boundaries. I looked over the go1 user manual as well as the software manual. I also looked over a little bit of the YOLO repository in order to see how we would be able to implement object detection in this project.

Software manual:

1.3 Units

In development, unspecified units are unified according to international standard units:

Length unit: meter (m)

Angle: radian (rad)

Angular velocity: radians per second (rad/s)

Torque: Nm (N.m)

Mass unit: kilograms (kg)

Inertial tensor unit: (kg-m²)

1.4 Coordinate, kinematics and dynamics

1.4.1 Joint number and joint limits

The quadruped robot is like an animal, and its trunk and legs are bilaterally symmetrical. The four legs are divided into two groups according to the front and back. The two groups are the same except for the front and rear. The coordinate system and joint motion range of the two groups are the same. But the coordinate system we defined is not mirror-symmetric, see section 1.4.2.

Number of legs and joints:

Leg0 FR = right front leg

Leg1 FL = left front leg

Leg2 RR = right rear leg

Leg3 RL = left rear leg

Joint 0: Hip, Hip joint

Joint 1: Thigh, Thigh joint

Joint 2: Calf, Calf joint

e.g. FR_thigh: right front leg thigh joint

Joint limitations:

Hip joint: $-60^{\circ} \sim 60^{\circ}$

Thigh joint: $-38^{\circ} \sim 170^{\circ}$

Calf joint: $-156^{\circ} \sim -48^{\circ}$

Go1

User Manual V1.4

2021.12



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go1_user_manual.pdf